

technology that is capable of providing all DSL services from one line card or home gateway. The aggregate downstream and upstream speed provided by UDSL is at least 200Mbps.

Ethernet local loop is a next-generation DSL technology that incorporates the features of Ethernet and DSL. It is capable of delivering speeds up to 6Mbps over a distance of 6.5km on a moderate-quality copper line. Very-high-speed DSL (VDSL) is an asymmetric technology that is capable of delivering downstream rates of 13Mbps to 52Mbps and upstream rates of 1.5Mbps to 2.3Mbps over a single copper pair line. The distance covered is from 300 metres to 1.4 kilometres from the telephone exchange.

Gigabit DSL (GDSL) technology is based on binder multiple-input and multiple-output (MIMO) technology and is capable of delivering 1Gbps up to 300-metre range.

Currently, the primary focus in xDSL wire-line technology is the development and deployment of ADSL technologies and architectures.

Asymmetric digital subscriber line (ADSL)

The Internet is used largely for downloading files, HTML and graphical content. Processes like uploading files or other content to servers are limited to very few users. Hence, the bandwidth required for downstream data is more than upstream data bandwidth requirement.

Further, on the technical side, it was realised that it was possible to transmit data more quickly from an exchange to a user, but that, when the user sent information to the exchange, it was more sensitive to noise caused by electromagnetic disturbances (nearer to the exchange, greater the concentration of cables, generating more cross-talk). Thus, the upload signal is weakest at the noisiest part of the local loop, while the download signal is strongest at the noisiest part of the local loop.

It therefore makes technical sense to have the digital subscriber line ac-

ADSL Versions Showing Up/Down Data Rates

ADSL version	Standard	Year	Upstream data rate	Downstream data rate
ADSL	ANSI T1.413-1998 issue 2	1998	1Mbps	8Mbps
ADSL G.DMT	ITU G.992.1	1999-07	1.3Mbps	12Mbps
ADSL over POTS	ITU G.992.1 annex A	2001	1.3Mbps	12Mbps
ADSL over ISDN	ITU G.992.1 annex B	2005	1.8Mbps	12Mbps
ADSL Lite	ITU G.992.2	1999-07	512kbps	1.5Mbps
ADSL2	ITU G.992.3	2002-07	1Mbps	12Mbps
ADSL2	ITU G.992.3 annex J	2002-07	3.5Mbps	12Mbps
RE-ADSL2	ITU G.992.3 annex L	2002-07	800kbps	5Mbps
Splitter-less ADSL2	ITU G.992.4	2002-07	500kbps	1.5Mbps
ADSL2+	ITU G.992.5	2003-05	1Mbps	24Mbps
ADSL2+M	ITU G.992.5 annex M	2008	3.5Mbps	24Mbps

cess multiplexer (DSLAM) transmit at a higher bit-rate than the modem on the customer end. So, the idea was to use an asymmetric system, imposing a lower speed from the subscriber to the exchange. This idea was incorporated by the employees of Bellcore (now Telcordia Technologies), who developed ADSL in 1988 by placing wide-band digital signals above existing voice signal on conventional twisted pair cables.

This DSL based technology enables transmission and reception of data at speeds higher than legacy copper media. There are two competing and incompatible standards for modulating the ADSL signal, known as discrete multi-tone (DMT) and carrierless amplitude phase (CAP).

CAP was the original technology used for DSL deployments, but the most-widely used method now is DMT. DMT modulation technique allows several bits to be represented by one transmission symbol. In ADSL, bit-rate allocation for a channel within the available bandwidth is not the same for other channels, and hence the term asymmetric.

In other words, the upstream bandwidth is smaller than the downstream bandwidth. ADSL offers an upstream data rate of 500kbps and a downstream data rate of up to 8Mbps.

ADSL-Lite, another variant of ADSL standard, offers upstream speeds up to 512kbps and down-

stream speeds of 1.5Mbps. Further, ADSL has many variants like ADSL2, splitter-less ADSL2, ADSL2+ and ADSL++.

ADSL standards

ADSL versions, along with their respective upstream and downstream data rates, are listed in Table above. ADSL2 or G.DMT.bis is defined in ITU G.992.3 and is an improved version of ADSL, with data rates in downstream as 12Mbps and 3.5Mbps in upstream. Splitter-less ADSL2 or G.lite.bis is defined in ITU G.992.4 and is capable of providing 1.536Mbps downstream and 512kbps upstream.

ADSL2+ defined in ITU G.992.5 can provide up to 24Mbps theoretical downstream speed, which is double the speed of ADSL2. The upstream speed is up to 3.5Mbps.

Thus, ADSL2+ doubles the frequency band of typical ADSL from 1.1MHz to 2.2MHz. More importantly, ADSL2+ provides port-bonding known as G.998.x or G.Bond. This is a very attractive feature of ADSL2+ in which the download and upload speeds is the sum of individual speed of all provisioned ports to the end-user. It means that, if two lines with 24Mbps were bonded, the net result would be a speed of 48Mbps.

ADSL++ , another variant of ADSL, developed by Centillium Communications, is capable of provid-